

西南财经大学本科期末考试试题册（A）

（2023—2024学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA SE）(英)

命题教师：王磊

适用对象（年级专业）：2023 信息管理与信息系统专业

使用试题的任课教师姓名：王磊

试题说明：

- 1、考试类型：闭卷[] 开卷[✓]
- 2、本套试题共 2 道大题，共 6 页，完卷时间 120 分钟。
- 3、考试用品中除纸、笔、尺子外，可另带的用具有：
计算器[] 字典[] 等
(请在下划线上填上具体数字或内容，所选[]内打钩)

考试时间（由制卷方填写）：

以下各项由学生填写：

任课教师：

年级专业：

学生姓名：

学 号：

- 考生注意事项：
- 1.出示学生证（或身份证）和准考证于桌面左上角，以备查验。
 - 2.拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。
 - 3.答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。
 - 4.所有答案均需填写在答题纸上，答在试题册上无效。
 - 5.考生不得携带任何通讯工具进入考场。
 - 6.考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。
 - 7.严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	2	3	4	5	6	7	8	9	10

1. A _____ is a program that executes compiled Java Bytecode on a specific platform.

- A. Java Virtual Machine
- B. Java Compiler
- C. Java Programming Manual
- D. Eclipse IDE

2. If you attempt to add an **int**, a **byte**, a **short**, and a **float**, the result will be a _____ value.

- A. int
- B. long
- C. byte
- D. float

3. Assuming x and y are both declared as integer variables and x = 17 and y = 6, which Java expression has the biggest value?

- A. $x / y + 3$
- B. $x \% y * 4$
- C. $x - y / 3$
- D. $x / 5 - y * 2$

4. Which of the following expressions is equivalent to the **boolean** expression:

$$!(A < 5 \ \&\& \ B \neq C)$$

- A. $A > 5 \ || \ B \neq C$
- B. $A \geq 5 \ \&\& \ B == C$

C. $!(A < 5) \parallel (B != C)$

D. $A >= 5 \parallel B == C$

5. What is the output from the following Java program fragment?

```
String S = ""; String T = "";  
int i = 4;  
for (i = 1; i <= 3; i++);  
    S = S + "!";  
for (i = 1; i < 4; i++)  
    T = T + "*";  
System.out.print(S);  
System.out.println(T);
```

A. (that is, the empty string, printed twice)

B. *

C. !***

D. !!!***

6. After executing the following Java code, what is output?

```
public class FinalExam {  
    public static void main(String[] args) {  
        String S1 = "Java Programming";  
        String S2 = " is taught";  
        String S3 = " at Colorado State";  
        int iSize = S3.length() + 3;  
        System.out.print(iSize);  
        char cChar = S1.charAt(10);  
        System.out.print(cChar);  
        int iIndex = S2.indexOf("p");  
        System.out.print(iIndex);  
    }  
}
```

```
}  
}
```

A. 21 r 5

B. 21 a -1

C. 21 a 5

D. 20 a 5

7. Which of these statements about constructors is wrong?

A. A constructor has no return type

B. Its name must be the same as the class in which it is defined

C. There must be exactly one constructor defined for a class

D. Constructors are almost always declared as public

8. An interface must meet which of these restrictions?

I. It must not have any fields.

II. All methods must be abstract.

III. Only public methods are allowed.

IV. A class can implement only one interface.

A. II and III only

B. II only

C. I, II and III only

D. II, III and IV only

9. Consider the following classes, which methods show an example of methods **overloading** and which methods show an example of methods **overriding**?

```
public class superClass {  
    public void method1(int x, int y){  
        System.out.println(x+" "+y);  
    }  
}
```

```

    int method2(){
        return 1;
    }
}
class subClass extends superClass{
    public void method1(){
        System.out.println("Hi!!");
    }
    public int method2(){
        return 2;
    }
}

```

- A. There are no overloaded methods, example of methods overriding: method2
- B. There are no overloaded methods, example of methods overriding: method1 and method2
- C. Example of methods overloading: method1, Example of methods overriding: method 2
- D. Example of methods overloading: method1, there are no overridden methods.

10. Which of the following statements is not true about **protected** data members?

- A. They can be accessed from the same class.
- B. They can be accessed from subclasses in different packages.
- C. They can be accessed from all other classes in the same package.
- D. None of the above.

Part II: Write Programs.(total 70 pts)

1. (10 pts) Suppose that a String object is only composed of letters and digits. Write a program that prompts the user to enter a String, and then displays the number of letters and digits respectively.

2. (15pts) In an array, numbers greater than 3 times its average are recognized as outliers. (e.g. 820 is recognized as an outlier in the below array)

```
int [ ] x= {7, 2, 13, 25, 1, 19, 33, 820, 5,41};
```

Complete the following method to display the total amount of outliers in above array.

```
public static int AmountOfOutliers(int [ ] a) {  
    }  
}
```

3. (20 pts) Design a **Printable** interface, which contains two abstract methods: `getArea( )` and `getPerimeter( )`, which returns the area and the perimeter of a geometry respectively.

Also, you should design an abstract **RegularPolygon** class, which describes polygons with sides having the same length. It contains:

- ✓ A member variable named **side_length** denotes the length of the regular polygon's side.
- ✓ A **constructor**.
- ✓ An abstract method, **degreeOfAngle()**, returns the degree of any angle.

Then, you should design a **Square** class, which extends the **RegularPolygon** class and implements the **Printable** interface at the same time. It at least contains the members below:

- ✓ A private member variable, **num_sides**, denotes the number of the polygon's sides.
- ✓ A constructor to initialize the member variables of the Square object.
- ✓ Overridden method.
- ✓ A **toString()** method to display basic information of the Square object, at least including its side length, degree of angle, area and perimeter.

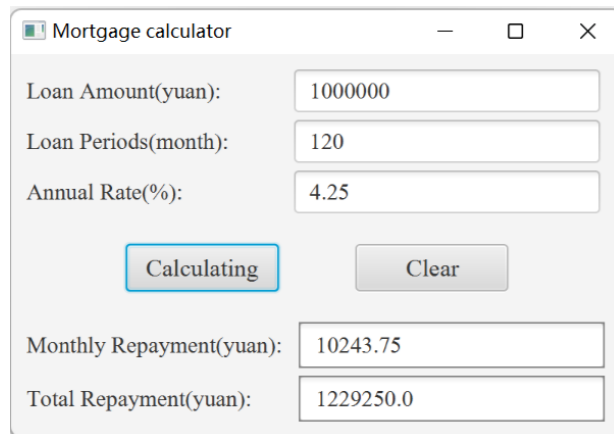
Finally, you should create a **Square** object with side length of 5 in the **main** method, and display its basic information.

4. (25 pts) Suppose a custom loaned 1,000,000 RMB from bank to purchase a house. He chose an equal installment method to repay the bank in n months. The monthly repayment is calculated by the below formula:

$$P = \frac{T \times (1 + i)^n \times i}{(1 + i)^n - 1}$$

Here, T denotes the total loan amount from the bank (e.g. $T=1,000,000$), i denotes the monthly rate (or annual rate divided by 12).

Now, please design a Java program to calculate monthly repayment for the custom, as shown in below figure.



Loan Amount(yuan):	1000000
Loan Periods(month):	120
Annual Rate(%):	4.25
Calculating Clear	
Monthly Repayment(yuan):	10243.75
Total Repayment(yuan):	1229250.0

Requirements:

- Use suitable Java FX GUI components.
- Use suitable layout to organize components.
- Contain at least one event handling process.